

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

First Semester of M. Tech. Mechanical (CAD/CAM) Examination

May-2013

ME 704.01 Advanced Machine Design

Date: 08.05.2013, Wednesday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.
5. PSG Design data book is allowed.

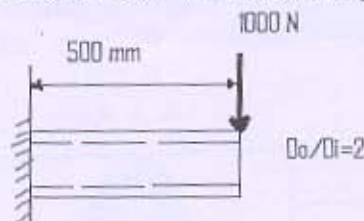
SECTION - I

Q - 1 (a) Discuss on various phases for a typical design process. [05]

Q - 1 (b) Explain following materials for application at elevated temperature. [07]

- Ferrite steel
- Stainless steel
- Cobalt and nickel base alloys
- Dispersion hardened based alloys

Q - 2 Design a spindle using strength and rigidity criterion. Consider $\sigma_t=70\text{MPa}$, $\tau=40\text{MPa}$, $\theta_{\max}=0.2^\circ$ per meter length and $\theta_{\max}$ for lateral rigidity $=0.3^\circ$ per meter length. Take $k_b=k_t=1.5$, $E=2.1 \times 10^5\text{MPa}$, $G=0.84 \times 10^5\text{MPa}$, Torque $=200000\text{Nmm}$. [10]



OR

Q - 2 Explain stress cycle (S-N curve) and state its importance to a designer with suitable case study. [10]

Q - 3 (a) Discuss on creep phenomena with necessary curve. [04]

Q - 3 (b) What is the importance of various theories of failures in the design of mechanical components? Explain the tresca criteria. [09]

OR

Q - 3 An edge crack beam, there is a crack in its central plane whose length is 5mm; a load of 2000N is applied opposite to crack, so that crack would tends to open in bending. Calculate the SIF of crack if the beam has the following dimensions: [13]

- Depth of the beam $w=25\text{mm}$
- Thickness $b=10\text{mm}$
- Span $S=100\text{mm}$

If two loads of 600 N each are placed at 25mm from central plane. What SIF of crack be?

SECTION – II

- Q - 4 Explain the importance of material selection and heat treatment in gear design. [08]
Q - 5 (a) What is interference? What step will you take to avoid interference? [06]
Q - 5 (b) Discuss the various loading patterns use for material handling system design. [08]

OR

- Q - 5 For S-Gearing determine the dimensions of gear keeping ration in center distance constant $m=7$, $\alpha=20^\circ$, $i=8.5$, $a=182$ mm, determine the diameter and contact ratio. [14]
Q - 6 (a) Discuss Human factor consideration for design of [03]
 - Lever of screw jack
 - Clutch lever of an automobile car.
 - Driver's seat in automobile truck.

Q - 6 (b) Design a tensile bar of length $L=200$ mm to carry a tensile load of 5 kN for minimum cost of the following materials: [10]

Material	Mass Density (kg/m ³)	Material Cost (Rs/N Weight)	Yield Strength (MPa)
1	7500	16	130
2	3000	32	50
3	4800	480	90
4	2100	32	20

OR

- Q - 6 (a) Comparison of concurrent engineering and traditional engineering. [03]
Q - 6 (b) Design a simple tensile bar of length L and load P with stiffness K_1 for cost optimization having constraints shown in figure below. [10]

$$K = K_1,$$
$$A \geq A_{min}$$

